# POSTAI (Context Engine)

## AI-Powered Social Media Marketing Assistant for Small Businesses

### Project Specification Document

**Submitted for**

**AMD AI DevMaster Hackathon 2026**

**Track 2 – Localized AI Agents Deployment**


Submitted By

**Sugitha Saravanan**

B.Tech – Artificial Intelligence & Data Science

V.S.B. Engineering College, Karur


GitHub Repository

**https://github.com/SUGITHA-A56/AI_Contact-_engine**

Demo Video

**https://drive.google.com/file/d/1zVBXp8w2Es9kMsxlz6pN37iq2YFLspv8/view**



# V.S.B. ENGINEERING COLLEGE

**(An Autonomous Institution)**

**(Anna University Recognized Research Institute)**

**Approved by AICTE, New Delhi, Affiliated to Anna University, Chennai.**

**NBA Accredited Courses, Accredited by NAAC**

## 1. INTRODUCTION

Small businesses in India — local boutiques, bakeries, tea shops, and jewellery stores — often struggle to maintain a consistent and effective social media presence. Creating engaging content, planning posting schedules, and keeping track of seasonal trends demands time and marketing expertise that most small business owners do not have. Hiring a dedicated marketing team or agency is often financially out of reach.

With the rapid growth of Artificial Intelligence, digital marketing solutions have become an effective way to provide small businesses with accessible, affordable, and personalized marketing support. PostAI (also referred to as the Context Engine) is an AI-powered web application developed to act as a digital marketing team for small businesses, helping them build a strong online presence without needing prior marketing knowledge.

The system stores each business's context — its products, brand voice, brand colors, and target audience — in a knowledge base, and uses this stored memory to generate relevant, on-brand social media content through the Claude API. It integrates a Context Engine (RAG-based business memory), a Strategy Planner, a Content Engine, and a Trend Calendar into one unified platform.

The frontend is developed using React.js, Vite, and Tailwind CSS, while the backend is built using Java Spring Boot with MySQL as the database. These technologies work together to create a responsive, secure, and efficient marketing assistant. Business owners set up their profile once, and the system keeps generating fresh, relevant, on-brand content over time, reducing repetitive manual effort.

The main objective of PostAI is to make professional-quality social media marketing accessible to small businesses by providing intelligent content generation, strategic planning, and festival-based trend awareness anytime and anywhere.

## 2. KEY FEATURES OF PostAI

PostAI is an AI-powered social media marketing assistant developed to provide reliable, accessible, and personalized marketing support for small businesses. The application integrates Artificial Intelligence with modern web technologies to improve marketing efficiency and business visibility. The key features of the system are as follows:

### 2.1 AI-Powered Content Engine

The application uses the Claude API to generate captions, hashtags, and creative content ideas. Business owners can request content on any theme, and the system produces on-brand, ready-to-post captions instantly.

### 2.2 Context Engine (RAG-Based Business Memory)

The Context Engine indexes the business profile and product catalog as vector embeddings. This indexed context is retrieved through Retrieval-Augmented Generation (RAG) to ensure every generated post accurately reflects the business's identity.

### 2.3 Strategy Planner

The Strategy Planner allows business owners to define content pillar ratios — Promotion, Education, Testimonials, and Offers — through an interactive donut chart, and view an auto-scheduled content calendar.

### 2.4 Knowledge Catalog

Products and services are added to a catalog with descriptions optimized for AI context retrieval, allowing the Content Engine to reference specific offerings accurately.

### 2.5 Trend Calendar

The Trend Calendar auto-discovers upcoming Indian festivals and local events relevant to the business's industry, along with recommended campaign ideas for each occasion.

### 2.6 Brand Aesthetics Configuration

Business owners can set their primary accent, secondary accent, and background colors, ensuring that generated creatives stay visually consistent with their brand identity.

### 2.7 Secure User Authentication

The application includes a secure registration and login system built with Java Spring Boot and MySQL, ensuring that only authorized users can access their business data.

### 2.8 Responsive and User-Friendly Interface

The frontend is developed using React.js, Vite, and Tailwind CSS, providing a clean, responsive interface that works efficiently across desktops, tablets, and mobile devices.

## 3. OBJECTIVES

The primary objective of PostAI is to develop an intelligent marketing assistant that provides reliable, on-brand social media content through Artificial Intelligence. The system aims to support small business owners by offering timely content generation, strategic planning, and easy access to essential marketing resources.

The specific objectives of the project are:

- To develop an AI-powered assistant that generates accurate and relevant social media content.
- To integrate the Claude API for intelligent, context-aware caption and hashtag generation.
- To build a Context Engine that stores business profile data as vector embeddings for RAG-based retrieval.
- To help business owners define and manage a balanced content strategy through the Strategy Planner.
- To maintain a Knowledge Catalog of products and services for accurate content recall.
- To provide a Trend Calendar that highlights relevant seasonal and festival marketing opportunities.
- To implement a secure user authentication system for protecting business data and personalized access.
- To develop a responsive and user-friendly web application using modern web technologies.
- To reduce the time and expertise required for small businesses to maintain a social media presence.
- To create a scalable platform that can be enhanced with additional marketing features in the future.

## 4. MODULES

PostAI is designed using a modular approach, where each module performs a specific function to ensure smooth and efficient operation of the system. The major modules are described below.

### 4.1 User Authentication Module

This module allows business owners to register and log in securely. It verifies user credentials and ensures that only authorized users can access the application's personalized features.

### 4.2 Business Profile (Context Engine) Module

This module lets business owners set up their business knowledge base — business name, industry, location, target audience, and brand colors — and index it for AI context retrieval.

### 4.3 Knowledge Catalog Module

This module manages the product/service catalog. Each entry includes a name, category, and AI-context description, stored as vector embeddings for accurate retrieval.

### 4.4 Strategy Planner Module

This module lets business owners define content pillar ratios and view a schedule showing when each type of content should be posted, along with a continuous feedback loop.

### 4.5 Content Engine Module

This is the core AI module. It uses the Claude API to generate captions, hashtags, and creative visual ideas based on the indexed business context and a user-provided prompt.

### 4.6 Trend Calendar Module

This module surfaces upcoming Indian festivals and local events relevant to the business's industry, along with recommended campaigns tailored to the business type.

### 4.7 Database and Backend Module

The backend is developed using Java Spring Boot, while MySQL is used for data storage. This module manages business data, authentication, and communication with the Claude API.

### 4.8 User Interface Module

The User Interface Module is developed using React.js, Vite, and Tailwind CSS. It provides a responsive, attractive, and easy-to-use interface across different devices.

## 5. SYSTEM ARCHITECTURE

The system architecture of PostAI represents the overall structure and interaction between different components of the application. The architecture follows a client-server model where the frontend, backend, AI service, and database work together to provide an efficient marketing assistant.

The Web Interface Layer acts as the entry point of the application. Users interact with the system through a responsive interface built using React.js, Vite, and Tailwind CSS. This layer provides access to the Business Profile, Strategy Planner, Content Engine, and Trend Calendar modules.

The Application Layer consists of the backend developed using Java Spring Boot. It manages request handling, authentication and authorization, business logic, API integration with the Claude API, and response management.

The AI Service Layer integrates the Claude API to provide intelligent, context-aware content generation. Business context retrieved through RAG is combined with the user's prompt to generate accurate, on-brand captions, hashtags, and strategy advice.

The Database Layer uses MySQL for secure storage and management of application data, including business profiles, product catalogs, and generated content history.

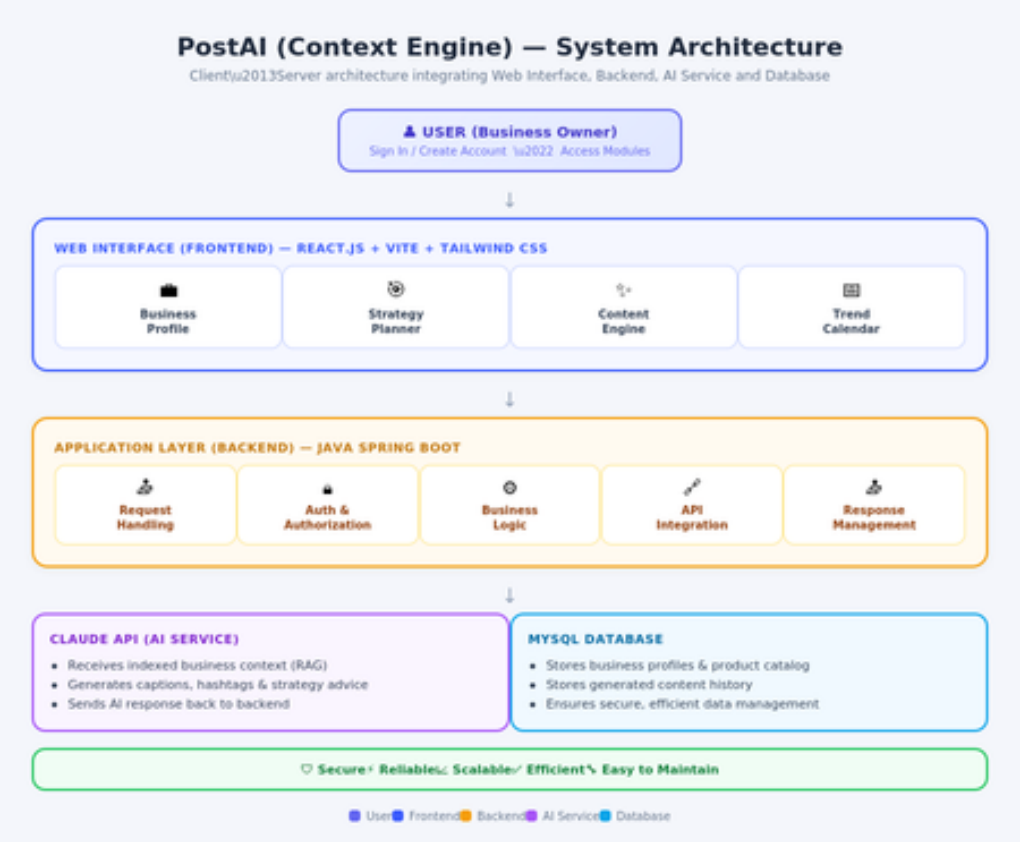



*Fig 5.1 – System Architecture of PostAI*

# 6. TECHNOLOGIES USED

PostAI is developed using various frontend, backend, database, and Artificial Intelligence technologies to provide an efficient and user-friendly marketing platform.

## 6.1 Frontend Technologies

React.js, Vite, and Tailwind CSS are used to design and develop the user interface of the application. These technologies provide a responsive, interactive, and user-friendly experience across different devices.

## 6.2 Backend Technology

Java Spring Boot is used for backend development. It manages business logic, REST APIs, user authentication, and communication between the frontend, database, and AI services.

## 6.3 Database Technology

MySQL is used for storing and managing business profiles, product catalogs, authentication data, and generated content history securely.

## 6.4 Artificial Intelligence Technology

The Claude API is integrated to provide AI-based content generation for social media captions, hashtags, and marketing strategy suggestions.

## 6.5 Development Tools

Visual Studio Code, MySQL Workbench, Postman, Git, and GitHub are used for coding, database management, API testing, and version control during development.



## V.S.B. ENGINEERING COLLEGE

**(An Autonomous Institution)**

**(Anna University Recognized Research Institute)**

**Approved by AICTE, New Delhi, Affiliated to Anna University, Chennai.**

**NBA Accredited Courses, Accredited by NAAC**

# 7. IMPLEMENTATION – CODING

The PostAI project is implemented using a combination of frontend, backend, database, and Artificial Intelligence technologies. The frontend is developed using React.js, Vite, and Tailwind CSS, while the backend is developed using Java Spring Boot with MySQL database integration. The following sections show the key implementation parts of the system.

## 7.1 User Authentication Implementation

The authentication module manages business owner registration and login. User details are validated and stored in the MySQL database through Spring Boot APIs.

User Registration API

```
@PostMapping("/api/register")

public ResponseEntity<?> registerUser(@RequestBody User user) {

    User savedUser = userService.saveUser(user);

    return ResponseEntity.ok(savedUser);

}
```

This function receives user details from the frontend and stores the registered user information securely in the database.

User Login Handler

```
@PostMapping("/api/login")

public ResponseEntity<?> loginUser(@RequestBody LoginRequest request) {

    User user = userService.validateUser(

        request.getUsername(),

        request.getPassword()

    );

    if (user != null) {

        return ResponseEntity.ok(user);

    }

    return ResponseEntity.status(401)

        .body("Invalid Credentials");

}
```

This API checks the entered username and password against stored database records and allows authenticated users to access the dashboard.

## 7.2 Business Profile (Context Engine) Implementation

The Business Profile module saves the business's knowledge base and triggers indexing for RAG-based retrieval.

```
@PostMapping("/api/business")

public ResponseEntity<?> saveBusinessProfile(

        @RequestBody BusinessProfile profile) {

    BusinessProfile saved =

        businessService.saveAndIndex(profile);

    return ResponseEntity.ok(saved);

}
```

This function receives the business name, industry, location, target audience, and brand colors, then stores and indexes them as vector embeddings.

## 7.3 Content Engine – Claude API Integration

The Content Engine module connects the application with the Claude API to generate captions and hashtags based on the retrieved business context.

```
@PostMapping("/api/content/generate")

public ResponseEntity<?> generateContent(

        @RequestBody ContentRequest request) {

    String context = contextEngine

        .retrieveContext(request.getBusinessId());

    String aiResponse = claudeApiService

        .generateCaption(context, request.getPrompt());

    return ResponseEntity.ok(aiResponse);

}
```

This API retrieves the indexed business context, sends it along with the user's prompt to the Claude API, and returns the generated caption.

## 7.4 Frontend – Content Generation Request

The frontend sends the user's prompt and business ID to the backend and displays the AI-generated response on the Content Engine page.

```
async function generateContent(prompt, businessId) {

  const response = await fetch("/api/content/generate", {

    method: "POST",

    headers: { "Content-Type": "application/json" },

    body: JSON.stringify({ prompt, businessId })

  });

  const data = await response.json();

  setGeneratedCaption(data.caption);

}
```

## 7.5 Knowledge Catalog – Add Product

This function adds a new product to the Knowledge Catalog, which is later indexed for RAG-based content generation.

```
@PostMapping("/api/products")

public ResponseEntity<?> addProduct(

        @RequestBody Product product) {

    Product saved = productService.save(product);

    return ResponseEntity.ok(saved);

}
```

### 7.6 Trend Calendar – Fetch Industry Trends

This function retrieves upcoming festival and trend data relevant to the business's industry for the Trend Calendar module.

```
@GetMapping("/api/trends/{industry}")

public ResponseEntity<?> getTrends(

        @PathVariable String industry) {

    List<Trend> trends =

        trendService.getTrendsByIndustry(industry);

    return ResponseEntity.ok(trends);

}
```

### 7.7 Database Connectivity

Spring Boot is connected with MySQL using JPA and Hibernate for managing business and content data.

```
spring.datasource.url=

    jdbc:mysql://localhost:3306/postai

spring.datasource.username=root

spring.datasource.password=password

spring.jpa.hibernate.ddl-auto=update
```

This configuration establishes communication between the backend application and the MySQL database.

# V.S.B. ENGINEERING COLLEGE

**(An Autonomous Institution)**

**(Anna University Recognized Research Institute)**

**Approved by AICTE, New Delhi, Affiliated to Anna University, Chennai.**

**NBA Accredited Courses, Accredited by NAAC**

## Entity Relationship (ER) Diagram

The ER diagram below illustrates how the User, Business, Post, and AI Engine entities relate to one another within the PostAI database.

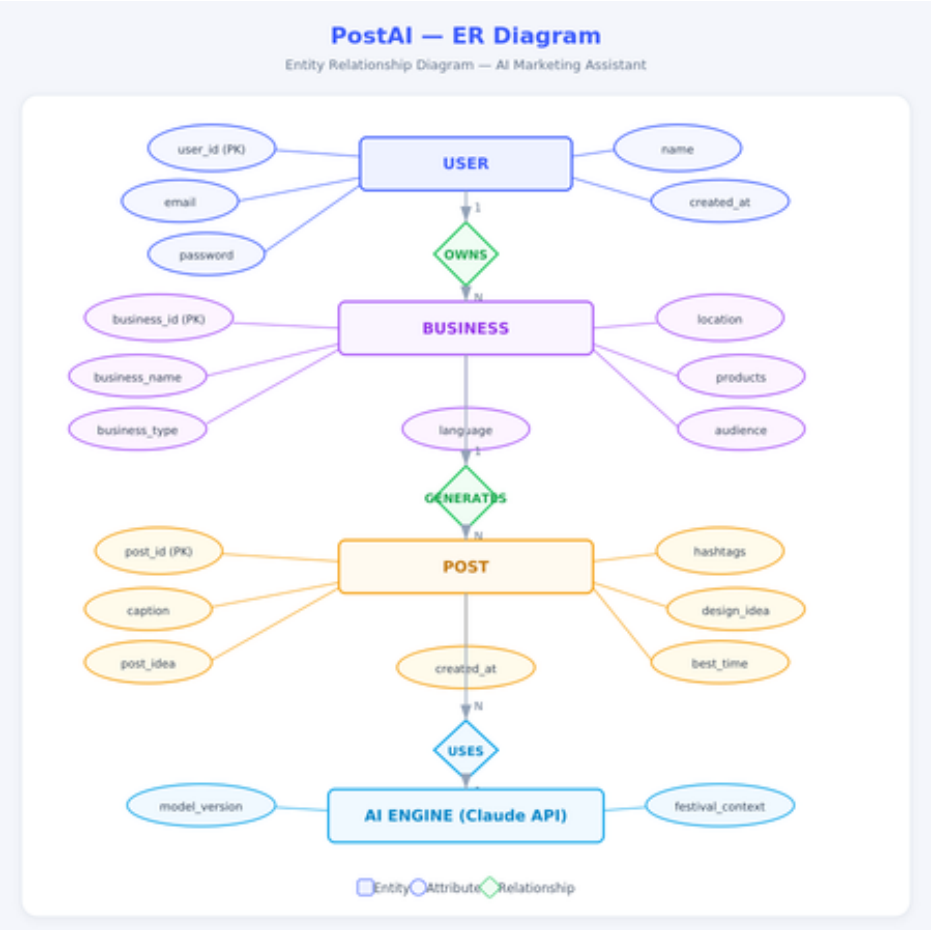



*Fig 7.1 – Entity Relationship Diagram*

# V.S.B. ENGINEERING COLLEGE

**(An Autonomous Institution)**

**(Anna University Recognized Research Institute)**

**Approved by AICTE, New Delhi, Affiliated to Anna University, Chennai.**

**NBA Accredited Courses, Accredited by NAAC**

## User Flow Diagram

The flow below shows the journey of a business owner from first opening the app through generating and reusing AI-powered content.

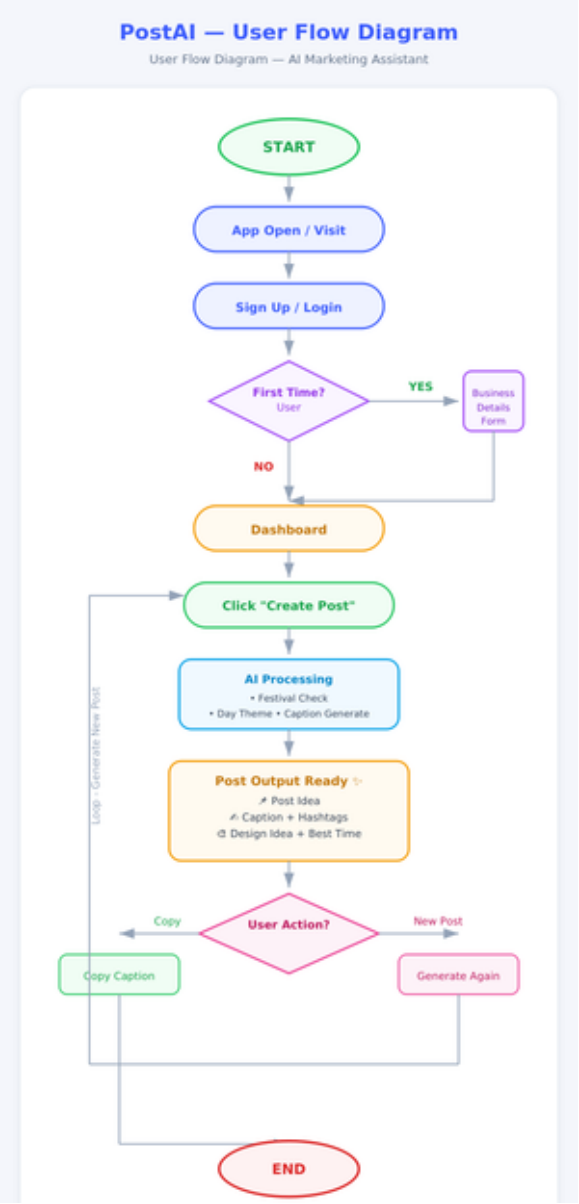



*Fig 7.2 – User Flow Diagram*

V.S.B. ENGINEERING COLLEGE

(An Autonomous Institution)

(Anna University Recognized Research Institute)

Approved by AICTE, New Delhi, Affiliated to Anna University, Chennai.

NBA Accredited Courses, Accredited by NAAC

# 8. OUTPUT SCREENSHOTS

The following screenshots demonstrate the working of PostAI.

## 8.1 Login / Landing Page

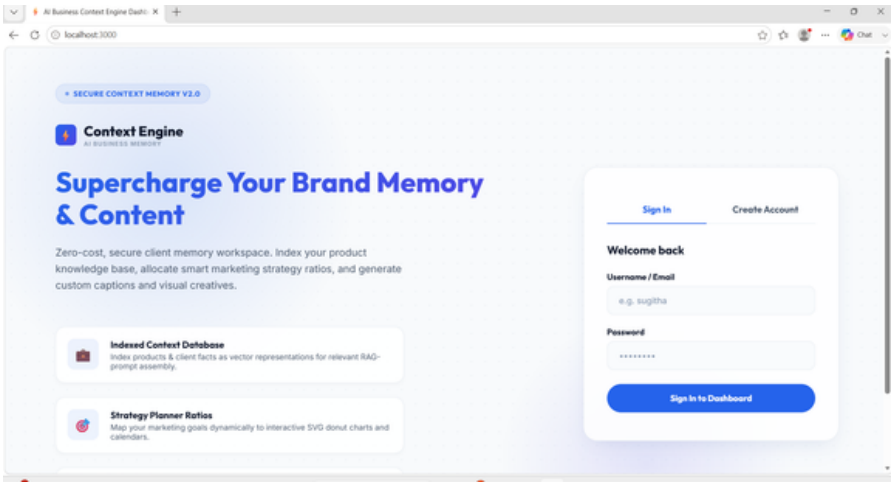



*Fig 8.1 – The landing page introduces the Context Engine platform, where a business owner can sign in or create a new account.*

## 8.2 Business Profile & Knowledge Catalog

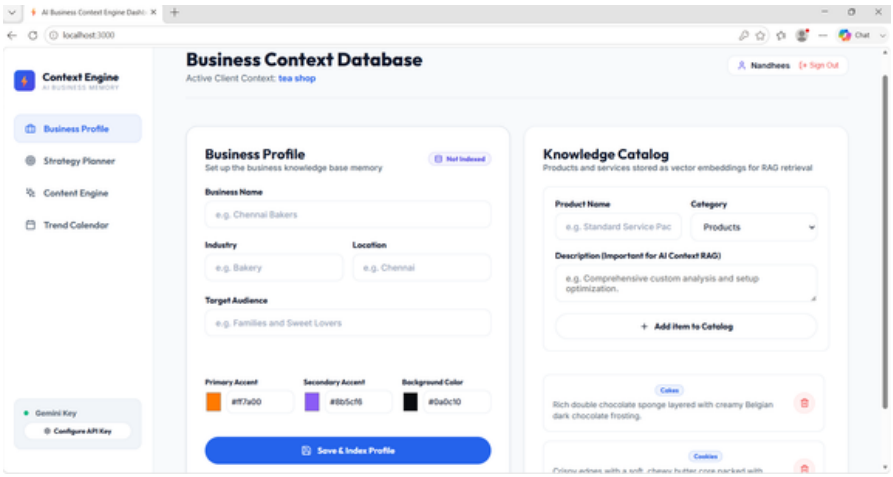



*Fig 8.2 – Business owners set up their profile (name, industry, location, target audience, brand colors) and manage the product catalog, indexed for RAG-based retrieval.*

## 8.3 Strategy Planner

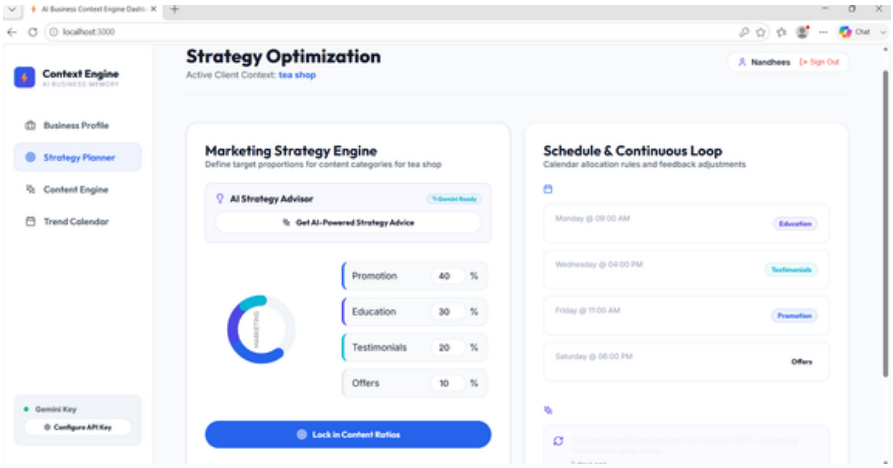



*Fig 8.3 – The Strategy Planner defines content pillar ratios (Promotion, Education, Testimonials, Offers) using an interactive donut chart.*

## 8.4 Content Engine

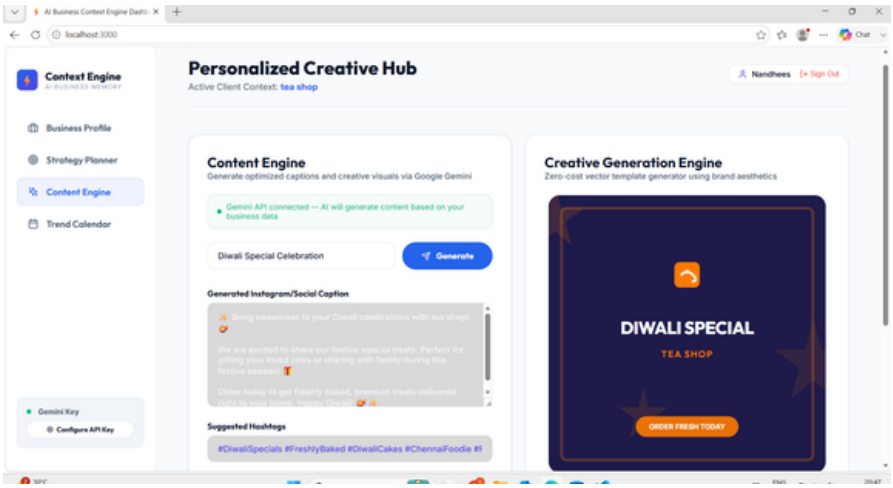



*Fig 8.4 – The Content Engine generates AI-powered captions, hashtags, and creative visuals based on the business's stored context.*

# V.S.B. ENGINEERING COLLEGE

**(An Autonomous Institution)**

**(Anna University Recognized Research Institute)**

**Approved by AICTE, New Delhi, Affiliated to Anna University, Chennai.**

**NBA Accredited Courses, Accredited by NAAC**

## 8.5 Trend Calendar – Business Context Intelligence

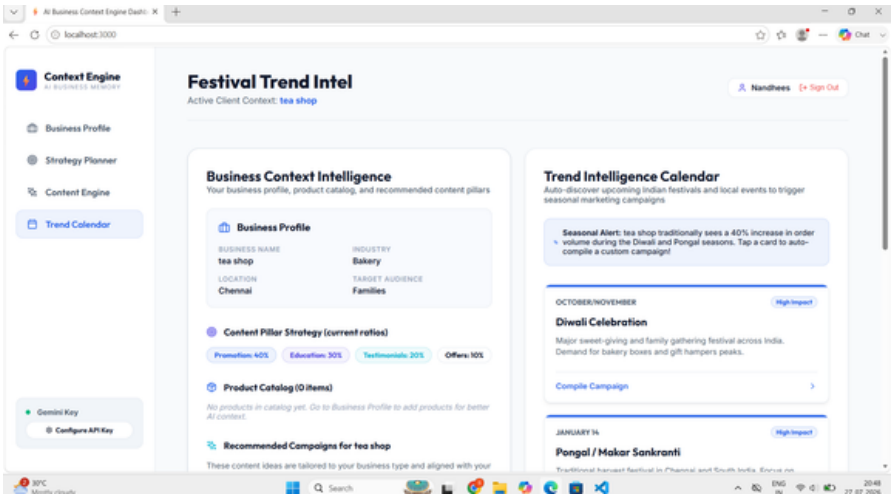



*Fig 8.5 – Displays the business profile summary, content pillar ratios, and upcoming high-impact festivals like Diwali and Pongal.*

## 8.6 Trend Calendar – Recommended Campaigns

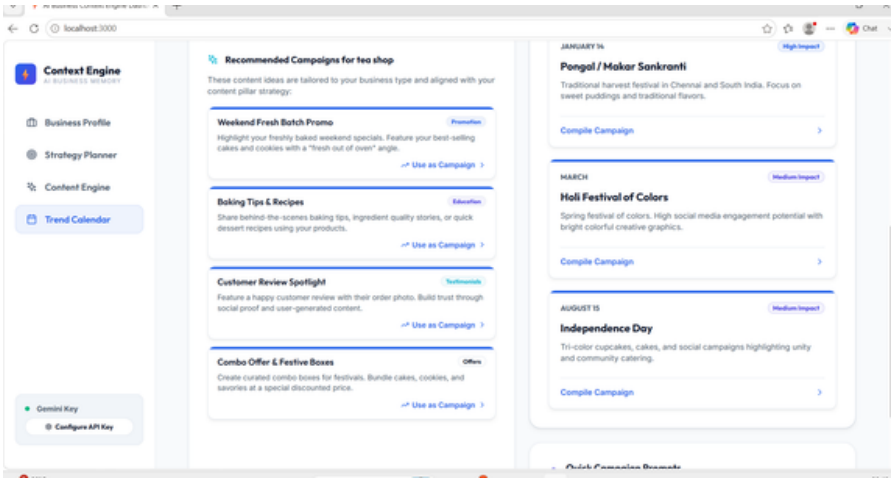



*Fig 8.6 – Shows campaign ideas tailored to the business type, along with a broader festival trend calendar including Holi and Independence Day.*

# 9. SOFTWARE USED

The development of PostAI involves the use of various software tools and platforms for designing, coding, testing, database management, and version control. The following software tools were used during the development process:

### 1. Visual Studio Code

Visual Studio Code is used as the primary code editor for developing the frontend and backend components of the application.

### 2. Spring Boot

Spring Boot is used for backend development. It helps in creating REST APIs, managing application logic, handling authentication, and connecting the application with the database.

### 3. MySQL Workbench

MySQL Workbench is used for designing, managing, and querying the MySQL database that stores business profiles, product catalogs, and content history.

### 4. Postman

Postman is used for testing and validating REST APIs, checking requests, responses, and backend functionality during development.

### 5. Git and GitHub

Git and GitHub are used for version control and source code management, maintaining project backups and tracking changes.

### 6. Web Browser (Google Chrome)

Google Chrome is used for running and testing the web application, verifying the user interface, responsiveness, and performance.

### 7. Claude API

The Claude API is used for integrating Artificial Intelligence capabilities into the application, enabling intelligent caption and content generation.



# V.S.B. ENGINEERING COLLEGE

**(An Autonomous Institution)**

**(Anna University Recognized Research Institute)**

**Approved by AICTE, New Delhi, Affiliated to Anna University, Chennai.**

**NBA Accredited Courses, Accredited by NAAC**

## 10. TECHNOLOGY STACK

The PostAI project is developed using a combination of frontend, backend, database, and Artificial Intelligence technologies. The technology stack provides a strong foundation for building a secure, responsive, and intelligent marketing platform.

### 1. Frontend Technologies

- React.js – Used to build reusable UI components across all modules.
- Vite – Used as the fast build tool and development server.
- Tailwind CSS – Used for styling and building a responsive, modern interface.

### 2. Backend Technologies

- Java – Used as the primary programming language for backend development.
- Spring Boot – Used to develop REST APIs, manage business logic, and handle authentication.
- Spring Data JPA / Hibernate – Used for object-relational mapping with the MySQL database.

### 3. Database Technology

- MySQL – Used for storing business profiles, product catalogs, and generated content securely.

### 4. Artificial Intelligence Technology

- Claude API – Integrated to provide AI-based conversational and content-generation capabilities for captions, hashtags, and strategy advice.

### 5. Development and Testing Tools

- Visual Studio Code – Used as the development environment for writing and managing source code.
- Postman – Used for testing and validating backend APIs.
- Git & GitHub – Used for version control and source code management.

# 11. ADVANTAGES, LIMITATIONS AND FUTURE ENHANCEMENTS

## 11.1 Advantages

- Easy Access to Marketing Guidance – Provides content ideas and captions anytime and anywhere, without depending on a marketing agency.
- AI-Based Personalized Support – The Claude API generates content tailored to each business's stored context, making marketing more relevant and efficient.
- Consistent Brand Voice – The Context Engine ensures every generated post reflects the business's name, industry, and target audience.
- Strategic Content Planning – The Strategy Planner helps businesses maintain a healthy balance of promotional, educational, and testimonial content.
- Festival-Aware Marketing – The Trend Calendar helps businesses capitalize on seasonal demand spikes like Diwali and Pongal.
- User-Friendly Interface – The responsive web design allows business owners to navigate different features efficiently.
- Secure Business Data Management – The authentication system ensures secure access to business accounts and protects business data.

## 11.2 Limitations

- Not a Replacement for a Marketing Team – The system provides content suggestions but cannot fully replace strategic human marketing expertise.
- Internet Dependency – The application requires an active internet connection to access the Claude API and generate content.
- Limited to Text and Basic Creatives – The current system focuses on captions, hashtags, and simple vector creatives rather than advanced video content.
- No Direct Social Media Publishing – The system does not yet auto-publish content directly to Instagram, Facebook, or other platforms.
- Language Limitations – The system currently has limited support for regional Indian languages.

## 11.3 Future Enhancements

- Multilingual Support – Adding regional Indian language support to help businesses reach local audiences more effectively.
- Automated Post Scheduling – Integrating direct scheduling and publishing to Instagram and Facebook.
- Analytics Dashboard – Adding engagement analytics to track how generated content performs over time.
- AI-Based Image Generation – Expanding the Creative Generation Engine with more advanced AI-generated visuals.
- Mobile Application Development – Converting the system into an Android/iOS application for better accessibility.
- Advanced Personalization – Learning from a business's past posting performance to refine future content suggestions.

## 12. CONCLUSION

PostAI successfully demonstrates the application of Artificial Intelligence and modern web technologies in the field of small business marketing. The system provides an intelligent and accessible platform that assists business owners by offering AI-generated content, strategic planning, product cataloging, and festival-based trend awareness.

The integration of the Claude API enables the system to generate accurate, on-brand captions and hashtags based on each business's stored context, while the Context Engine's RAG-based retrieval ensures relevance and consistency. The combination of React.js, Vite, Tailwind CSS, Java Spring Boot, and MySQL helps in developing a secure, responsive, and efficient marketing application.

PostAI helps reduce the gap between small businesses and professional-quality digital marketing by providing instant, relevant content assistance anytime and anywhere. The application encourages consistent brand presence, supports data-driven content strategy, and acts as a digital marketing companion for business owners.

Although the system does not replace a full marketing team, it serves as a valuable and affordable tool that provides meaningful support. With future enhancements such as multilingual support, automated scheduling, analytics, and AI-based image generation, PostAI can be further developed into a comprehensive marketing solution for small businesses across India.

In conclusion, PostAI represents an innovative approach to using Artificial Intelligence for improving small business marketing accessibility, helping business owners build and maintain a strong social media presence with confidence.

**GitHub Link:** https://github.com/SUGITHA-A56/AI_Contact-_engine

**Project Demo Video:** https://drive.google.com/file/d/1zVBXp8w2Es9kMsxlz6pN37iq2YFLspv8/view?usp=sharing

## THANK YOU!!